Recall: A subset $S \subseteq F$ is **algebraically independent** over $K \subseteq F$ if for all $s_1, \ldots, s_n \in S$ and $f(x_1, \ldots, x_n) \in K[x_1, \ldots, x_n]$ such that if $f(s_1, \ldots, s_n) = 0$, then $f = 0$.

$S \subseteq F$ is a **transcendence basis** for F over K if S is a maximal algebraically independent set.

Theorem: If S is a finite transcendence basis for F/K and T is another transcendence basis, then $|S| = |T|$.

Proof: Let $S = \{s_1, \ldots, s_n\}$ with s_i distinct and let T be another transcendence basis. Suppose there is no $t \in T$ such that t is transcendental over $K(s_2, \ldots, s_n)$. Then all elements of T are algebraic over $K(s_2, \ldots, s_n)$. So $K(s_2, \ldots, s_n, T)$ is an algebraic extension of $K(s_2, \ldots, s_n)$. But F is an algebraic extension of $K(T) \subseteq K(s_2, \ldots, s_n, T)$ so F is algebraic over $K(s_2, \ldots, s_n)$, but s_1 is algebraic over $K(s_2, \ldots, s_n)$, so S was not algebraically independent. So there exists $t_1 \in T$ that is transcendental over $K(s_2 \ldots s_n)$ so $\{t_1, s_2, \ldots, s_n\}$ is algebraically independent.

Claim: s_1 is algebraic over $K(t_1, s_2, \ldots, s_n)$.

Otherwise, s_1 is transcendental over $K(t_1, s_2, \ldots, s_n)$ but $\{s_1, t_1, \ldots, s_n\}$ is algebraically independent which contradicts the maximality of $S = \{s_1, \ldots, s_n\}$

So F is an algebraic extension of $K(s_1, \ldots, s_n)$, so it is an algebraic extension of $K(s_1, t_1, \dots, s_n)$ which is an algebraic extension of $K(t_1, s_2, \dots, s_n)$, so $\{t_1, s_2, \dots, s_n\}$ is a transcendence basis for F/K, so $\{t_1, \dots, t_n\}$ is a transcendence basis for some $\{t_1, \dots, t_n\} \subseteq T$, then $\{t_1, \dots, t_n\}$ is maximally algebraically independent, so $T = \{t_1, \dots, t_n\}$.

Remark: $\operatorname{trdim}(F/K)$ is well-defined, and also, if $F/E/K$, $\operatorname{trdim}(F/K) = \operatorname{trdim}(F/E) + \operatorname{trdim}(E/K)$.

Commutative Algebras

Henceforth, all rings are commutative rings with identity.

Definition: An R-module A is Noetherian if any sequence of submodules $A_1 \subseteq A_2 \subseteq \cdots \subseteq A$ is eventually constant, that is, there exists $N \in \mathbb{N}$ such that $A_n = A$ for all $n \geq N$.

An R-module A is Artinian if any sequence of submodules $A_1 \supseteq A_2 \supseteq \cdots \supseteq A_n$ is eventually constant.

Example: $\mathbb{Z}$ is Noetherian. Any ideal is principal, so $\langle m_1 \rangle \subseteq \langle m_2 \rangle \subseteq \cdots$ if and only if $m_n \mid m_{n+1}$ $\forall n \in \mathbb{N}$

However, $\mathbb{Z}$ is not Artinian. Take $A_n = \langle p^n \rangle$

Theorem: A module is Noetherian if and only if any non empty collection of submodules has a maximal element.

Proof: If $A_1 \subseteq A_2 \subseteq \cdots$, then has a maximal element.

Conversely, if some nonempty collection S has no maximal element. Choose $A_1 \in S$, then $A_2 \in S$ with $A_1 \subsetneq A_2$, and proceed inductively.

Theorem: A module is Noetherian if and only if every submodule is finitely generated.

Proof: Let A be a R-module and M be a submodule. Assume M is not finitely generated. Then for any $m_1 \in M$ $M \neq \langle m_1 \rangle$. Choose $m_2 \in M \setminus \langle m_1 \rangle$ such that $M \neq \langle m_1, m_2 \rangle$. Proceed inductively,

$\langle m_1 \rangle \subseteq \langle m_1, m_2 \rangle \subseteq \cdots$ does not stabilize, so not Noetherian.

Conversely, if all submodules of A are finitely generated and $A_1 \subseteq A_2 \subseteq \cdots \subseteq A$ be submodules. Then

$A_\infty = \bigcup_{n=1}^{\infty} A_n$ is a submodule of A, so

$A_\infty = \langle a_1, \dots, a_m \rangle$ for some $a_1, \dots, a_m \in A_\infty$, so

$a_1, \dots, a_m \in \bigcup_{n=1}^{N} A_n = A_N$. Then for $n \geq N$, $A_N \subseteq A_n \subseteq A_\infty$.

Theorem: Let $0 \to A \to B \to C \to 0$ be an exact sequence of R-modules. Then B is Noetherian if and only if A and C are

Proof: If M is Noetherian and $N \subseteq M$ is a submodule, then N is Noetherian because submodules of N $\{N_n\}_{n=1}^{\infty}$ such that $N_n \subseteq N_{n+1}$ $\forall n \in \mathbb{N}$ are also submodules of M. So if

$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ is exact, and B is Noetherian, $f(A) \subseteq B$ is Noetherian and $A \simeq f(A)$.

If $C_1 \subseteq C_2 \subseteq \cdots \subseteq C$, then $g^{-1}(C_1) \subseteq g^{-1}(C_2) \subseteq \cdots \subseteq B$ so there is a $N \in \mathbb{N}$ such that $g^{-1}(C_N) = g^{-1}(C_n)$ $\forall N \geq n$, so $C_N = C_n$ for all $N \geq n$, so C is Noetherian.

Conversely, if A and C are Noetherian, let $B_1 \subseteq B_2 \subseteq \cdots \subseteq B$ be a sequence of submodules. For $n \in \mathbb{N}$, let $C_n = g(B_n)$, so $C_1 \subseteq \cdots \subseteq C$, let $A_n = f^{-1}(B_n) = f^{-1}(f(A) \cap B)$, so $A_1 \subseteq \cdots \subseteq A$. There exists $N \in \mathbb{N}$ such that $A_N = A_n$ and $C_N = C_n$ $\forall N \geq n$.

We have exact sequences

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
0 \arrow[r] & A_n \arrow[r, "f"] \arrow[d, "id"] & B_n \arrow[r, "g"] \arrow[d, "id"] & C_n \arrow[r] \arrow[d, "id"] & 0 \\
0 \arrow[r] & A_N \arrow[r, "f"] \arrow[d] & B_N \arrow[r, "g"] & C_N \arrow[r] \arrow[d] & 0 \\
 & 0 & & 0 &
\end{tikzcd}$$

Claim: $B_N = B_n$

Let $y \in B_N$ Then there is $x \in B_n$ such that $g(x) = g(y)$. Then $y - x \in \ker(g) \cap B_N = \mathrm{Im}(f) \cap B_N = f(A) \cap B_N$
$A_N = f^{-1}(f(A) \cap B_N)$ so $\exists z \in A_N$ s.t. $f(z) = y - x$. But $A_N = A_n$, so $f(z) \in B_n$, so $y = x + f(z) \in B_n$.

So B is Noetherian.

Corollary: If $A_1, \ldots, A_n$ are R-modules, then $\bigoplus_{i=1}^{n} A_i$ is Noetherian iff, $\forall 1 \leq k \leq n$, A_k are.

Proof: $0 \to A_1 \to A_1 \oplus A_2 \to A_2 \to 0$ and induction.

So a finitely generated R-module $A \simeq R^n$ is Noetherian if R is.

Theorem: Any finitely generated module over a Noetherian ring is Noetherian.